I decided to use everything—to the last bolt and bearing—that SEP used in Viking, the French equivalent of Vikas.

It was not so much because I did not trust those parts we had developed—just that I did not want to give any of my detractors to start the chorus against liquid propulsion again. If Vikas failed the test, a failure analysis committee will pinpoint the reason, which would mostly be associated with a part. Then I could always argue that the part was the same that is tested and used by SEP, and that their engines are working. We could replace the part and test again.

Having mounted the engine on to the test stand, every day we would load the propellant into the tank and await the wind, in the right direction. For a few days this continued, the conditions not being favourable. By 3:30 pm the propellant will be unloaded. This exercise would take an hour and a half, and by 5 pm the French scientists would religiously close shop and go home.

As we awaited the test, ISRO chemical engineer P Sudarshan, who had earlier opposed to the Vernon test, made a visit. Seeing the struggle we were going through, he wrote back to ISRO an appreciation letter on our efforts.

'I hadn't quite understood this,' he told me.

Finally, there came the perfect day. There was strong wind blowing from the direction of Vernon town. It was so cold that we had to wear multiple layers of clothing. It was 12 December 1985, my 44th birthday.

Vikas stood on the stand, ready to be tested. Along an arc a few hundred meters away, there was the instrumentation room, a control room and a store. We were to sit in one of the rooms. During the many tests I have witnessed at PF2, the test stand in Vernon, scientists would sit in one of the rooms, peering into consoles, monitoring the performance of various parameters of the engine being tested. I preferred to stand in the open, something that the security protocol does not allow, and gaze at the fire that the engine spewed, listening to the roar of machine that man had made to send his creations to yonder skies. The trembling of the ground as the test stand held back the engine that the liquid propulsion sought to lift was pure